INTELLIGENT DATA CLEANING & VALIDATION BOT

Enterprise Deployment Overview

Version 1.0

1. Executive Summary

Organizations routinely handle Excel / CSV datasets that are incomplete, inconsistent, and error-prone. Manual cleaning is slow, non-scalable, and audit-invisible.

The Intelligent Data Cleaning & Validation Bot is an enterprise-grade, containerized system that:

- Standardizes Excel datasets using deterministic cleaning logic
- Generates a structured audit log for flagged data
- Prevents duplicate file processing while allowing controlled re-processing after a job has been completed. Re-processing is allowed; however, all attempts to process same file are counted and logged.
- Supports concurrent users
- Provides secure API-based access
- Preserves data privacy (no third-party transmission)

The system is designed for internal enterprise deployment and controlled integration.

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2. Problem Statement

Operational datasets commonly contain:

- Mixed date formats
- Misspellings and abbreviations
- Invalid numeric tokens
- Ambiguous coordinate structures
- Duplicate entries
- Missing values

Traditional Excel-based cleaning:

- Is manual
- Is inconsistent
- Has no audit trace
- Does not scale across teams
- Consumes significant operational time that could otherwise be used for analytical tasks.

This bot transforms raw Excel files into analytics-ready structured datasets while preserving traceability.

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3. System Architecture

3.1 Technology Stack

- FastAPI – API Layer
- PostgreSQL – Durable metadata storage (users + jobs)
- Redis – Queue management, locks, rate limits
- RQ (Redis Queue) – Background job execution
- Docker – Containerized deployment
- SQLAlchemy – ORM for DB layer

3.2 Workflow

1. User authentication (JWT-based login)
2. File upload (.xlsx , .csv)
3. Deterministic Job-ID generation (HMAC-SHA256 based on file content)
4. Atomic claim lock (prevents duplicate uploads under concurrency)
5. Background worker processes file.
6. Status endpoint returns:
    a. queued
    b. processing
    c. done
    d. failed
7. Download endpoints:
    a. cleaned file
    b. logs file
8. Job is deleted only after both files are downloaded

All file processing occurs within the enterprise infrastructure.

No external data transfer occurs.

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3.3 Idempotency & Concurrency Control

- Job-ID derived from file content (not filename)
- Atomic Redis claim lock (SET NX EX)
- One-active-job-per-user enforcement (DB row lock)
- Atomic file publish via `os.replace()` (no partial files exposed as only the complete final file is written to the final file path)

This ensures correctness even under concurrent upload attempts.

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4. Cleaning Logic (Deterministic Rules)

The bot uses rule-based deterministic transformations:

Date Handling

- Standard format: YYYY-MM-DD
- Range: 1970–2035
- Ambiguous dates flagged
- Invalid numeric structures flagged
- Month-first date formats (MM-DD-YYYY) are not interpreted. In ambiguous cases such as `02-03-2024`, the system assumes the day-first format (DD-MM-YYYY), which aligns with common Indian date conventions
- Flagged values returned in log
- Date Intelligence switch has been provided for allowing flexibility regarding date cleaning strictness.
- Date tokens containing alphabetic characters within numeric positions (e.g., `2C-08-2024`) are flagged instead of cleaned, as the alphabetic character may represent any digit, making deterministic correction unreliable.

- 2-letter month abbreviations and months with wrong first letters are flagged to maintain strict cleaning logic

Location Validation

- Controlled location master list (Considered Indian State and UT names, official abbreviation of state and UT names, city names – Delhi, Mumbai, Kanpur, Chandigarh). Rest of the names have been ignored to show that the bot can identify and clean based on targeted names.
- Location matching based on: ≥80% similarity threshold
- Abbreviation matching
- Multiple locations flagged [2 locations under flagged multiple locations are displayed for convenience]; state-state, abbreviations-state considered multiple locations if present in a single cell of either of location column.
- Ambiguous names flagged
- Original value preserved and flagged values inserted in logs.

Coordinate Validation

- Enforced DMS format (latitude, longitude format)
- Requires N/S/E/W poles for clear data and prevent ambiguity.
- Minutes and seconds values exceeding 60 are corrected using carry-forward normalization.
- Out-of-bounds flagged. (>90 degree for latitude, >180 degree for longitude)
- Ambiguous coordinate tokens flagged

Price Normalization

- Standard format: Rs. number
- Decimal preservation
- Lakhs/crores/Thousands converted to numeric
- Ranges of price flagged
- Negative values converted to positive

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Duplicate Handling

- Duplicate rows removed post-cleaning
- Original row-id of duplicate and the row it matched to post cleaning recorded in logs

5. Outputs Generated

For each job:

1. cleaned_data.xlsx (or .csv) (cleaned file included with flags)
2. logs_data.xlsx (or .csv) (contains full flagged rows with flag messages + row-id)

6. Performance Characteristics

Tested on:

- 4 CPU cores
- 8GB RAM
- 3 workers stable

Performance:

- ~15–16 seconds for 10,000 rows
- ~4–5 seconds for 100 rows

Actual throughput depends on:

- Worker count
- Hardware capacity
- File complexity (File on which it was tested on had errors in almost all rows)

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7. Security & Access Control

- Password hashing (bcrypt mechanism)
- Authentication is implemented using signed JWT bearer tokens with configurable expiry, and all authenticated requests are expected to be transmitted only over enterprise-managed HTTPS channels.
- Ownership validation per job
- Rate limits:
    - Login endpoint (Rate limit=5 login attempts in 10 minutes, configurable)
    - Status polling (Rate limit=60 in 1 minute, configurable)
- Max file size enforced (configurable – set in environment variable file, 25MB default)
- Job timeout enforced (configurable, 10 minutes for a single processing as maximum time allotted to a worker exceeding which kills the process and marks the status as failed)
- All uploads are validated before processing.
- File paths are generated internally and not user supplied.
- Database stores information about file posting user, file details, file paths, timestamps, attempts - how many times the file was posted, Configuration version number of the program used, version hash for configuration integrity verification.

No third-party API calls.
Data remains within enterprise infrastructure.

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8. Deployment & Integration Model

8.1 Delivery Model

- Delivered as Docker image
- No direct access to enterprise infrastructure
- Source code sharing optional (based on agreement)

8.2 Enterprise Responsibilities

Enterprise IT will:

- Host container internally
- Provide Redis instance
- Provide PostgreSQL instance
- Configure environment variables
- Enable HTTPS via internal gateway/reverse proxy
- Deployment orchestration, including Kubernetes, container runtime configuration, networking, ingress, and internal infrastructure integration, will be handled entirely by the enterprise's IT team.

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8.3 System Requirements (Recommended)

- 4 CPU cores
- 8GB RAM
- SSD recommended
- Redis persistence recommended but not mandatory as PostgreSQL maintains durable job metadata.
- The above requirements enable 3 workers in Redis to work simultaneously. Increasing the number of workers requires a proportionally higher CPU and memory resources.
- System supports horizontal scaling by increasing worker instances.

8.4 Required Services

- Redis (queue + locks)
- PostgreSQL (durable metadata storage)

9. Configuration Layer

The system supports configuration-driven rules:

- Year range configurable
- Date intelligence switch
- Version Number
- Similarity threshold configurable
- Validation toggles configurable
- Flag messages
- Location list
- Price format and pattern matching
- Job timeout
- Atomic claim key expiry time
- Job key expiry time in Redis
- Maximum file size allowed
- Rate limit attempts and time limit for it
- Thresholds adjustable via configuration without code modification.
- All configuration changes require controlled deployment. Some configurations require changes in the environment variable file.

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10. SLA & Operational Policies

- Job timeout: 10 minutes
- Failed jobs marked explicitly
- Jobs retained until both outputs downloaded
- Automatic cleanup post-download
- Duplicate uploads blocked while a job is active

11. Data Retention Policy

- Input, cleaned, and log files stored temporarily
- Deleted after both outputs downloaded
- No long-term file storage

Job metadata retained in PostgreSQL for audit.

12. Limitations

- Structured tabular data (Excel or CSV) expected
- No machine-learning-based inference (deterministic system)
- Config change requires redeployment

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13. Strategic Positioning

This system is not a prompt-based AI tool.

It is a:

- Deterministic data standardization engine
- Workflow-controlled system
- Concurrency-safe processing framework
- Audit-preserving enterprise automation layer

Designed for reliability, traceability, and scalability.

SAMPLE OUTPUTS:

RAW DATA:

	A	B	C	D
1	location_to	location_from	Coordinate_to	Coordinate_from
2	Near Oddisha - Noid a- Branch	RNACHAL PRADESH (HYJDERABAD) @	20°28'33"Nz | 84°1'14"ENA	28°1'47"n 95°24'9"e
3	Chennai, Ladakh - Branch	bv andaman and nicobar islands (ahmedaba)d !!	34°19'333"N,77°31'40"E	10°59'2 N"9, 9323'26"E
4	OPP. GOA. -- GURUGRAM - BR.	Andhra Pradhse - Bengaluru	15°24'29"n , 74°58'13"e	15°54'2"N,80°15'19N"E
5	lcknww, gujarat ##	Ranchi La hkad- DC	21°'740°27;N'E	34°47'10n,79°8'33"e
6		noid, chdnaigarh		13°0'N | 76 EE'05° N/A
7	Andhra Pradesh - Bhubaneswar	Jaipur, Maharashtra (HQ.)	15°32'58"N,'94°9712"E	19°52'25"n,75°2'58"e
8	Behind Ahmedabad Andhra Pradesh - DC	jupiar, delhi	15°10'"6N,9°23'29"E	29°8'31"N,77°52'16"E
9	Skkim - Ranchi	Delhi - Jaipur	27°50'0"N,8°81'16"E N/A	27°44'56N",77°°9'49"E
10	Tamil Nadu (Hamedabad)	ABOVE JHARKHAND - PUNE.	not available	24°15'N,83°5D0'E
11	ChennaitUttar Pradesh - BR	Andaman AND. Nicobar Islands (Ranchi)	26°57OO42"N,82°29'9"E | NA	11°45'111"n;92°40'47"
12	ABOVE WEST BENGAL - AHMEDABAD	OdIsHa (RaCnI)	22°56'5"n ,8 9°27'3"	20°21'47"N,85°24'24"E
13	Grgram, GOA. - WH.	Jipur, Assam, Bharat	1°52'38"N 73°34'50"E	25°34'Nt;93°4q5'E
14	Gurugram, Uttarakkhand	Nagaland (Lhucknow) - Branch	28°53'N778°40'E	25°47'N,9v°26e
15		LADAKH (MUMBAI), IN	(22°20'8"N | 87°57q'43"E)	34°25'N | 755°2'E
16	HMACHAL PRADESH (NAGPUUR)		32°22'2i"n , 7'65°620"E	23°35'48"n | 88°6'26"e
17	mabumi uttarakhqand, in	Punev Mizoram, Bharat	30°21'30"N;779°5'8"	23°38'4"4N,91°47'51"E
18	MUMBAI MEGHALAYA	Near Ranc ,ihArunachal Pradesh (HQ)	25°0'N , 91°54'E	28°21'N | N94?E'2°?
19	behind pune darda and nagar haveli and daman and diu - dc		20°41'59"N;72°10'30"E	23°42'20"N , 92°38'12"E —A NA
20	sikkkim - hyderabad	bhubaneswar goa	27°45'16"N | 88°19'55"E	(15°2'13"N,72°K5E"94'0)
21	Mizoramm(Bhubaneswar) - Br	Mnpr (Ahmedabad)	23°57m25"N 91°e4'1"E	23°54'41"nM | 94°57'6"E
22	Goa - Beulagnru	BENGALURU, TELANGANA	14°5X'14"n | 73°23'49"E	
23	Puen Mizoram	chnnai, gujarat, bharat	23°57'45"n , 90°59'56"e nna	22°53'33"n,72°39'p2"e
24	Nagaland (Punne)	hydrbad, nagalnad	24°19'N,95°29'E	26°36'55"n,94°9'0"E
25		Pn pDuChErRy iN @	1634'53"n,-1811'6"	11°58'N | 80°9'E
26	Bihar (Bhubaneawar)		24°44'54JN,85°31'2U"E	15°22'11"N 851'2°1"E
27	RANCHI HARYANA	kerala (pnue) - wh	30°25'46"j , 75°26'1"	10°32'n | 76°59'e
28	Arunachal Pradesh - Gurugram, IN	near meghalaya - nagpur	27°33'21"n , 93°39'8"E	25°47'36"N , 92°1'2"E
29	Gurugram Mizordm - DC	Above Mumbaiq Lakshadweep	22°48'28"N |V92°"7'93E	10°4'55"N 71°299'53"e

	E	F
1	price	date
2	880.00	2U/10/2025
3	929	14--edc-2025
4	1.35	18-AUG-2025 | N/A
5	827.56	04/11/x2025
6	₹ 844	202-07-11
7	1.86	23--FeB-2025
8	2193.21	2025-03-12
9	1.447	20225-07-19
10	-754.44	
11	286	04-AU-20I24
12	1562.47	15/03/2025
13	986	202-601-04
14	1607.66	14Oct-2t52
15	717	26/12/025
16	449	23/10/0225
17	₹ 2	02//06/2025
18	1206.72	
19		
20	2.515	2gua-6-q024
21	2 INR	05-dEc-2025
22	1725	27-MAR-025
23	557.17	02/12/2024
24	755.92	2S024-10-06 | NA
25	-2441 INR	28-nove2025
26	45972.5	01/05/2025 / n
27	68600,00	13/I8/2024
28	1.267	2025-08-03
29	1029.16	202-509-08

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CLEANED FILE PRODUCED:

	A	B	C
1	row-id	location_to	location_from
2	1	Odisha	Arunachal Pradesh
3	2	Ladakh	Andaman and Nicobar Islands
4	3	[FLAG] Abbreviation - State conflict: OPP. GOA. -- GURUGRAM - BR. -> BR, Goa	Andhra Pradesh
5	4	Gujarat	Ladakh
6	5	[FLAG] No Location	Chandigarh
7	6	Andhra Pradesh	Maharashtra
8	7	Andhra Pradesh	Delhi
9	8	Sikkim	Delhi
10	9	Tamil Nadu	Jharkhand
11	10	[FLAG] Abbreviation - State conflict: ChennaitUttar Pradesh - BR -> BR, Uttar Pradesh	Andaman and Nicobar Islands
12	11	West Bengal	Odisha
13	12	Goa	Assam
14	13	Uttarakhand	Nagaland
15	14	[FLAG] No Location	[FLAG] LADAKH (MUMBAI), IN Row: 14, Multiple locations found: Mumbai (100%), Ladakh (100%)
16	15	Himachal Pradesh	[FLAG] No Location
17	16	Uttarakhand	Mizoram
18	17	[FLAG] MUMBAI MEGHALAYA Row: 17, Multiple locations found: Mumbai (100%), Meghalaya (100%)	Arunachal Pradesh
19	18	Dadra and Nagar Haveli and Daman and Diu	[FLAG] No Location
20	19	Sikkim	Goa
21	20	[FLAG] Abbreviation - State conflict: Mizoramm(Bhubaneswar) - Br -> BR, Mizoram	[FLAG] Could not find match location: Mnpr (Ahmedabad) , Similarity score:72.72727272727273%
22	21	Goa	Telangana
23	22	Mizoram	Gujarat
24	23	Nagaland	Nagaland
25	24	[FLAG] No Location	Puducherry
26	25	Bihar	[FLAG] No Location
27	26	Haryana	Kerala
28	27	Arunachal Pradesh	Meghalaya
29	28	Mizoram	[FLAG] Above Mumbaiq Lakshadweep Row: 28, Multiple locations found: Mumbai (100%), Lakshadweep (100%

	D	E	F
1	Coordinate_to	Coordinate_from	price
2	[FLAG] Ambiguous pole: 20°28'33"NZ | 84°1'14"ENA	28.0°1.0'47.0"N,95.0°24.0'9.0"E	Rs.880
3	34.0°24.0'33.0"N,77.0°31.0'40.0"E	[FLAG] Missing coordinates: 10°59'2 N"9, 9323'26"E	Rs.929
4	15.0°24.0'29.0"N,74.0°58.0'13.0"E	[FLAG] Malformed coordinate: 15°54'2"N,80°15'19N"E	Rs.1.35
5	[FLAG] Malformed coordinate: 21°'740°27;N'E	[FLAG] Malformed coordinate: 34°47'10N,79°8'33"E	Rs.827.56
6	[FLAG] Missing coordinates: NAN	[FLAG] Ambiguous pole: 13°0'N | 76 EE'05° N/A	Rs.844
7	[FLAG] Missing coordinates: 15°32'58"N,'94°9712"E	19.0°52.0'25.0"N,75.0°2.0'58.0"E	Rs.1.86
8	[FLAG] Malformed coordinate: 15°10'"6N,9°23'29"E	29.0°8.0'31.0"N,77.0°52.0'16.0"E	Rs.2193.21
9	27.0°50.0'0.0"N,9.0°21.0'16.0"E	[FLAG] Malformed coordinate: 27°44'56N",77°°9'49"E	Rs.1.447
10	[FLAG] Missing coordinates: NOT AVAILABLE	[FLAG] Ambiguous coordinates words between digits: 24°15'N,83°5D0'E	Rs.754.44
11	[FLAG] Ambiguous coordinates words between digits: 26°57OO42"N,82°29'9"E | NA	[FLAG] Invalid coordinates : Multiple or no E or W found: 11°45'111"N;92°40'47"	Rs.286
12	[FLAG] Malformed coordinate: 22°56'5"N ,8 9°27'3"	20.0°21.0'47.0"N,85.0°24.0'24.0"E	Rs.1562.47
13	1.0°52.0'38.0"N,73.0°34.0'50.0"E	[FLAG] Ambiguous pole: 25°34'NT;93°4Q5'E	Rs.986
14	[FLAG] Out of Bound Longitude: 28°53'N778°40'E	[FLAG] Ambiguous coordinates words between digits: 25°47'N,9V°26E	Rs.1607.66
15	[FLAG] Ambiguous coordinates words between digits: (22°20'8"N | 87°57Q'43"E)	[FLAG] Out of Bound Longitude: 34°25'N | 755°2'E	Rs.717
16	[FLAG] Ambiguous coordinates words between digits: 32°22'2I"N , 7'65°620"E	23.0°35.0'48.0"N,88.0°6.0'26.0"E	Rs.449
17	[FLAG] Out of Bound Longitude: 30°21'30"N;779°5'8"	[FLAG] Malformed coordinate: 23°38'4"4N,91°47'51"E	Rs.2
18	25.0°0.0'N,91.0°54.0'E	[FLAG] Malformed coordinate: 28°21'N | N94?E'2°?	Rs.1206.72
19	20.0°41.0'59.0"N,72.0°10.0'30.0"E	23.0°42.0'20.0"N,92.0°38.0'12.0"E	[FLAG]-No Price
20	27.0°45.0'16.0"N,88.0°19.0'55.0"E	[FLAG] Ambiguous coordinates words between digits: (15°2'13"N,72°K5E"94'0)	Rs.2.515
21	[FLAG] Ambiguous coordinates words between digits: 23°57M25"N 91°E4'1"E	[FLAG] Ambiguous pole: 23°54'41"NM | 94°57'6"E	Rs.2
22	[FLAG] Ambiguous coordinates words between digits: 14°5X'14"N | 73°23'49"E	[FLAG] Missing coordinates: NAN	Rs.1725
23	23.0°57.0'45.0"N,90.0°59.0'56.0"E	[FLAG] Ambiguous coordinates words between digits: 22°53'33"N,72°39'P2"E	Rs.557.17
24	24.0°19.0'N,95.0°29.0'E	26.0°36.0'55.0"N,94.0°9.0'0.0"E	Rs.755.92
25	[FLAG] Missing coordinates: 1634'53"N,°1811'6"	11.0°58.0'N,80.0°9.0'E	Rs.2441
26	[FLAG] Ambiguous pole: 24°44'54JN,85°31'2U"E	15.0°22.0'11.0"N,16.0°11.0'1.0"E	Rs.45972.5
27	[FLAG] Invalid coordinates - missing N/S/E/W: 30°25'46"J , 75°26'1"	10.0°32.0'N,76.0°59.0'E	Rs.6860000
28	27.0°33.0'21.0"N,93.0°39.0'8.0"E	25.0°47.0'36.0"N,92.0°1.0'2.0"E	Rs.1.267
29	[FLAG] Ambiguous coordinates words between digits: 22°48'28"N |V92°"7'93E	10.0°4.0'55.0"N,75.0°59.0'53.0"E	Rs.1029.16

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	E	F	G
1	a	price	date
2		Rs.880	[FLAG] INVALID DATE couldn't parse: 2U/10/2025
3		Rs.929	[FLAG] No month found 14--edc-2025
4		Rs.1.35	2025-08-18
5		Rs.827.56	2025-11-04
6		Rs.844	[FLAG] INVALID DATE couldn't parse: 202-07-11
7		Rs.1.86	2025-02-23
8		Rs.2193.21	2025-03-12
9		Rs.1.447	[FLAG] INVALID DATE: 20225-07-19
10		Rs.754.44	[FLAG] No Date
11		Rs.286	[FLAG] Invalid numeric structure 04-AU-20I24
12		Rs.1562.47	2025-03-15
13		Rs.986	2026-01-04
14		Rs.1607.66	[FLAG] Invalid numeric structure 14Oct-2t52
15		Rs.717	[FLAG] INVALID DATE couldn't parse: 26/12/025
16		Rs.449	[FLAG] INVALID DATE: 23/10/0225
17		Rs.2	2025-06-02
18		Rs.1206.72	[FLAG] No Date
19		[FLAG]-No Price	[FLAG] No Date
20		Rs.2.515	[FLAG] Invalid numeric structure 2gua-6-q024
21		Rs.2	2025-12-05
22		Rs.1725	2025-03-27
23		Rs.557.17	2024-12-02
24		Rs.755.92	2024-10-06
25		Rs.2441	2025-11-28
26		Rs.45972.5	2025-05-01
27		Rs.6860000	[FLAG] INVALID DATE couldn't parse: 13/I8/2024
28		Rs.1.267	2025-08-03
29		Rs.1029.16	2025-09-08

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LOGS FILE PRODUCED:

	A	B	C
1	row_id	location_to	location_from
2	1	Odisha	Arunachal Pradesh
3	2	Ladakh	Andaman and Nicobar Islands
4	3	[FLAG] Abbreviation - State conflict: OPP. GOA. -- GURUGRAM - BR. -> BR, Goa	Andhra Pradesh
5	4	Gujarat	Ladakh
6	5	[FLAG] No Location	Chandigarh
7	6	Andhra Pradesh	Maharashtra
8	7	Andhra Pradesh	Delhi
9	8	Sikkim	Delhi
10	9	Tamil Nadu	Jharkhand
11	10	[FLAG] Abbreviation - State conflict: ChennaitUttar Pradesh - BR -> BR, Uttar Pradesh	Andaman and Nicobar Islands
12	11	West Bengal	Odisha
13	12	Goa	Assam
14	13	Uttarakhand	Nagaland
15	14	[FLAG] No Location	[FLAG] LADAKH (MUMBAI), IN Row: 14, Multiple locations found: Mumbai (100%), Ladakh (100%)
16	15	Himachal Pradesh	[FLAG] No Location
17	16	Uttarakhand	Mizoram
18	17	[FLAG] MUMBAI MEGHALAYA Row: 17, Multiple locations found: Mumbai (100%), Meghalaya (100%)	Arunachal Pradesh
19	18	Dadra and Nagar Haveli and Daman and Diu	[FLAG] No Location
20	19	Sikkim	Goa
21	20	[FLAG] Abbreviation - State conflict: Mizoramm(Bhubaneswar) - Br -> BR, Mizoram	[FLAG] Could not find match location: Mnpr (Ahmedabad) , Similarity score:72.727272727272273%
22	21	Goa	Telangana
23	22	Mizoram	Gujarat
24	24	[FLAG] No Location	Puducherry
25	25	Bihar	[FLAG] No Location
26	26	Haryana	Kerala
27	28	Mizoram	[FLAG] Above Mumbaiq Lakshadweep Row: 28, Multiple locations found: Mumbai (100%), Lakshadweep (100%)

	D	E
1	Coordinate_to	Coordinate_from
2	[FLAG] Ambiguous pole: 20°28'33"NZ | 84°1'14"ENA	28.0°1.0'47.0"N,95.0°24.0'9.0"E
3	34.0°24.0'33.0"N,77.0°31.0'40.0"E	[FLAG] Missing coordinates: 10°59'2 N"9, 9323'26"E
4	15.0°24.0'29.0"N,74.0°58.0'13.0"E	[FLAG] Malformed coordinate: 15°54'2"N,80°15'19N"E
5	[FLAG] Malformed coordinate: 21°'740°27;N'E	[FLAG] Malformed coordinate: 34°47'10N,79°8'33"E
6	[FLAG] Missing coordinates: NAN	[FLAG] Ambiguous pole: 13°0'N | 76 EE'05° N/A
7	[FLAG] Missing coordinates: 15°32'58"N,'94°9712"E	19.0°52.0'25.0"N,75.0°2.0'58.0"E
8	[FLAG] Malformed coordinate: 15°10'"6N,9°23'29"E	29.0°8.0'31.0"N,77.0°52.0'16.0"E
9	27.0°50.0'0.0"N,9.0°21.0'16.0"E	[FLAG] Malformed coordinate: 27°44'56N",77°°9'49"E
10	[FLAG] Missing coordinates: NOT AVAILABLE	[FLAG] Ambiguous coordinates words between digits: 24°15'N,83°5D0'E
11	[FLAG] Ambiguous coordinates words between digits: 26°57OO42"N,82°29'9"E | NA	[FLAG] Invalid coordinates : Multiple or no E or W found: 11°45'111"N;92°40'47"
12	[FLAG] Malformed coordinate: 22°56'5"N ,8 9°27'3"	20.0°21.0'47.0"N,85.0°24.0'24.0"E
13	1.0°52.0'38.0"N,73.0°34.0'50.0"E	[FLAG] Ambiguous pole: 25°34'NT;93°4Q5'E
14	[FLAG] Out of Bound Longitude: 28°53'N778°40'E	[FLAG] Ambiguous coordinates words between digits: 25°47'N,9V°26E
15	[FLAG] Ambiguous coordinates words between digits: (22°20'8"N | 87°57Q'43"E)	[FLAG] Out of Bound Longitude: 34°25'N | 755°2'E
16	[FLAG] Ambiguous coordinates words between digits: 32°22'2I"N , 7'65°620"E	23.0°35.0'48.0"N,88.0°6.0'26.0"E
17	[FLAG] Out of Bound Longitude: 30°21'30"N;779°5'8"	[FLAG] Malformed coordinate: 23°38'4"4N,91°47'51"E
18	25.0°0.0'N,91.0°54.0'E	[FLAG] Malformed coordinate: 28°21'N | N94?E'2°?
19	20.0°41.0'59.0"N,72.0°10.0'30.0"E	23.0°42.0'20.0"N,92.0°38.0'12.0"E
20	27.0°45.0'16.0"N,88.0°19.0'55.0"E	[FLAG] Ambiguous coordinates words between digits: (15°2'13"N,72°K5E"94'0)
21	[FLAG] Ambiguous coordinates words between digits: 23°57M25"N 91°E4'1"E	[FLAG] Ambiguous pole: 23°54'41"NM | 94°57'6"E
22	[FLAG] Ambiguous coordinates words between digits: 14°5X'14"N | 73°23'49"E	[FLAG] Missing coordinates: NAN
23	23.0°57.0'45.0"N,90.0°59.0'56.0"E	[FLAG] Ambiguous coordinates words between digits: 22°53'33"N,72°39'P2"E
24	[FLAG] Missing coordinates: 1634'53"N,°1811'6"	11.0°58.0'N,80.0°9.0'E
25	[FLAG] Ambiguous pole: 24°44'54JN,85°31'2U"E	15.0°22.0'11.0"N,16.0°11.0'1.0"E
26	[FLAG] Invalid coordinates - missing N/S/E/W: 30°25'46"J , 75°26'1"	10.0°32.0'N,76.0°59.0'E
27	[FLAG] Ambiguous coordinates words between digits: 22°48'28"N |V92°"7'93E	10.0°4.0'55.0"N,75.0°59.0'53.0"E

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	F	G	H
1	price	date	Duplicate Rows
2	Rs.880	[FLAG] INVALID DATE couldn't parse: 2U/10/2025	Duplicate rows: {101, 102, 103}. Matched with rows: [62, 101],[56, 102],[95, 103]
3	Rs.929	[FLAG] No month found 14--edc-2025	Duplicate rows: {101, 102, 103}. Matched with rows: [62, 101],[56, 102],[95, 103]
4	Rs.1.35	2025-08-18	Duplicate rows: {101, 102, 103}. Matched with rows: [62, 101],[56, 102],[95, 103]
5	Rs.827.56	2025-11-04	Duplicate rows: {101, 102, 103}. Matched with rows: [62, 101],[56, 102],[95, 103]
6	Rs.844	[FLAG] INVALID DATE couldn't parse: 202-07-11	Duplicate rows: {101, 102, 103}. Matched with rows: [62, 101],[56, 102],[95, 103]
7	Rs.1.86	2025-02-23	Duplicate rows: {101, 102, 103}. Matched with rows: [62, 101],[56, 102],[95, 103]
8	Rs.2193.21	2025-03-12	Duplicate rows: {101, 102, 103}. Matched with rows: [62, 101],[56, 102],[95, 103]
9	Rs.1.447	[FLAG] INVALID DATE: 20225-07-19	Duplicate rows: {101, 102, 103}. Matched with rows: [62, 101],[56, 102],[95, 103]
10	Rs.754.44	[FLAG] No Date	Duplicate rows: {101, 102, 103}. Matched with rows: [62, 101],[56, 102],[95, 103]
11	Rs.286	[FLAG] Invalid numeric structure 04-AU-20I24	Duplicate rows: {101, 102, 103}. Matched with rows: [62, 101],[56, 102],[95, 103]
12	Rs.1562.47	2025-03-15	Duplicate rows: {101, 102, 103}. Matched with rows: [62, 101],[56, 102],[95, 103]
13	Rs.986	2026-01-04	Duplicate rows: {101, 102, 103}. Matched with rows: [62, 101],[56, 102],[95, 103]
14	Rs.1607.66	[FLAG] Invalid numeric structure 14Oct-2t52	Duplicate rows: {101, 102, 103}. Matched with rows: [62, 101],[56, 102],[95, 103]
15	Rs.717	[FLAG] INVALID DATE couldn't parse: 26/12/025	Duplicate rows: {101, 102, 103}. Matched with rows: [62, 101],[56, 102],[95, 103]
16	Rs.449	[FLAG] INVALID DATE: 23/10/0225	Duplicate rows: {101, 102, 103}. Matched with rows: [62, 101],[56, 102],[95, 103]
17	Rs.2	2025-06-02	Duplicate rows: {101, 102, 103}. Matched with rows: [62, 101],[56, 102],[95, 103]
18	Rs.1206.72	[FLAG] No Date	Duplicate rows: {101, 102, 103}. Matched with rows: [62, 101],[56, 102],[95, 103]
19	[FLAG]-No Price	[FLAG] No Date	Duplicate rows: {101, 102, 103}. Matched with rows: [62, 101],[56, 102],[95, 103]
20	Rs.2.515	[FLAG] Invalid numeric structure 2gua-6-q024	Duplicate rows: {101, 102, 103}. Matched with rows: [62, 101],[56, 102],[95, 103]
21	Rs.2	2025-12-05	Duplicate rows: {101, 102, 103}. Matched with rows: [62, 101],[56, 102],[95, 103]
22	Rs.1725	2025-03-27	Duplicate rows: {101, 102, 103}. Matched with rows: [62, 101],[56, 102],[95, 103]
23	Rs.557.17	2024-12-02	Duplicate rows: {101, 102, 103}. Matched with rows: [62, 101],[56, 102],[95, 103]
24	Rs.2441	2025-11-28	Duplicate rows: {101, 102, 103}. Matched with rows: [62, 101],[56, 102],[95, 103]
25	Rs.45972.5	2025-05-01	Duplicate rows: {101, 102, 103}. Matched with rows: [62, 101],[56, 102],[95, 103]
26	Rs.6860000	[FLAG] INVALID DATE couldn't parse: 13/l8/2024	Duplicate rows: {101, 102, 103}. Matched with rows: [62, 101],[56, 102],[95, 103]
27	Rs.1029.16	2025-09-08	Duplicate rows: {101, 102, 103}. Matched with rows: [62, 101],[56, 102],[95, 103]

Duplicate rows list both the duplicate record and the original row it matches.

Row-id+1=Excel Index [In original file]

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SAMPLE OUTPUT FOR DATE WHEN DATE INTELLIGENCE SWITCH IS TURNED OFF (FOR SAME SAMPLE ROWS OF INPUT PROVIDED ABOVE):

	F	G
1	price	date
2	Rs.880	[FLAG] INVALID DATE couldn't parse: 2U/10/2025
3	Rs.929	[FLAG] INVALID DATE couldn't parse: 14--edc-2025
4	Rs.1.35	[FLAG] INVALID DATE couldn't parse: 18-AUG-2025 | N/A
5	Rs.827.56	[FLAG] INVALID DATE couldn't parse: 04/11/x2025
6	Rs.844	[FLAG] INVALID DATE couldn't parse: 202-07-11
7	Rs.1.86	[FLAG] INVALID DATE couldn't parse: 23--FeB-2025
8	Rs.2193.21	2025-03-12
9	Rs.1.447	[FLAG] INVALID DATE couldn't parse: 20225-07-19
10	Rs.754.44	[FLAG] No Date
11	Rs.286	[FLAG] INVALID DATE couldn't parse: 04-AU-20I24
12	Rs.1562.47	2025-03-15
13	Rs.986	[FLAG] INVALID DATE couldn't parse: 202-601-04
14	Rs.1607.66	[FLAG] INVALID DATE couldn't parse: 14Oct-2t52
15	Rs.717	[FLAG] INVALID DATE couldn't parse: 26/12/025
16	Rs.449	[FLAG] INVALID DATE couldn't parse: 23/10/0225
17	Rs.2	2025-06-02
18	Rs.1206.72	[FLAG] No Date
19	[FLAG]-No Price	[FLAG] No Date
20	Rs.2.515	[FLAG] INVALID DATE couldn't parse: 2gua-6-q024
21	Rs.2	[FLAG] INVALID DATE couldn't parse: 05-dEc-2025
22	Rs.1725	[FLAG] INVALID DATE couldn't parse: 27-MAR-025
23	Rs.557.17	2024-12-02
24	Rs.755.92	[FLAG] INVALID DATE couldn't parse: 2S024-10-06 | NA
25	Rs.2441	[FLAG] INVALID DATE couldn't parse: 28-nove2025
26	Rs.45972.5	[FLAG] INVALID DATE couldn't parse: 01/05/2025 / n
27	Rs.6860000	[FLAG] INVALID DATE couldn't parse: 13/I8/2024
28	Rs.1.267	2025-08-03
29	Rs.1029.16	[FLAG] INVALID DATE couldn't parse: 202-509-08

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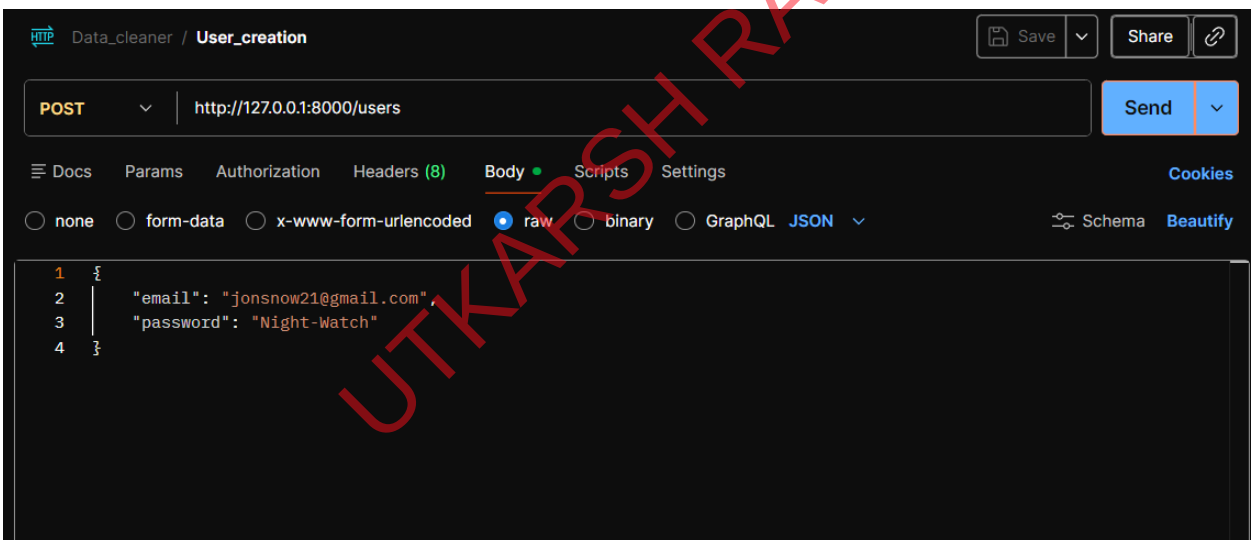
Month abbreviations were not cleaned rather flagged, any format of date which is broken and / or has any extra words or noise has not been cleaned rather flagged for targeted human intervention.

Following images of Postman are shared to highlight the mechanism of accessing the data cleaner:

USER CREATION:



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USER LOGIN WITH CORRECT PASSWORD

Data_cleaner / **User_login**

Save | Share

POST http://127.0.0.1:8000/login **Send**

Docs | Params | Authorization | Headers (8) | Body | Scripts | Settings | Cookies

none | form-data | x-www-form-urlencoded | raw | binary | GraphQL

	Key		Value	Description	Bulk Edit
☑	username	Text	jonsnow21@gmail.com		
☑	password	Text	Night-Watch		
	Key	Text	Value	Description	

Body | Cookies | Headers (4) | Test Results

200 OK • 2.20 s • 288 B • Save Response

JSON | Preview | Visualize

```
1 {
2   "access_token": [redacted] YOUR ACCESS TOKEN [redacted]
3   "token_type": "bearer"
4 }
```

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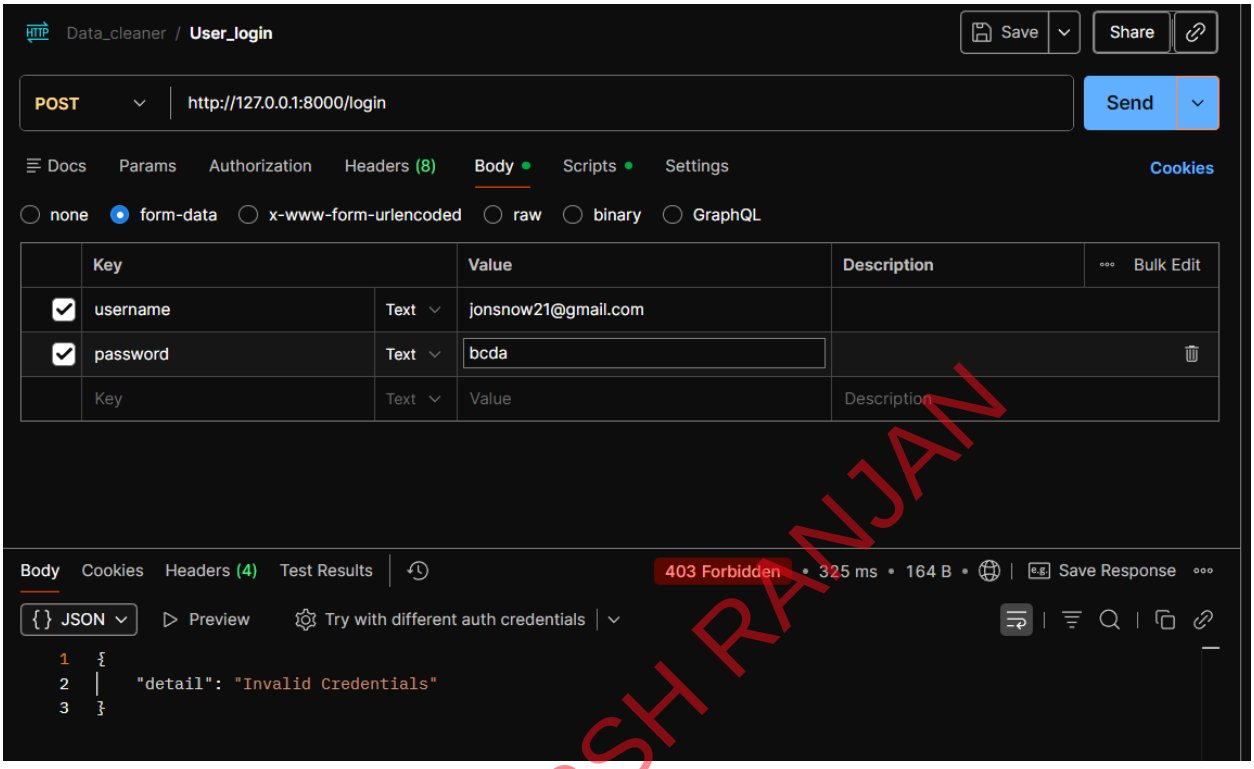
USER LOGIN WITH WRONG PASSWORD



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POSTING FILE

Data_cleaner / **Post_file**

Save | Share

POST http://127.0.0.1:8000/posts/ **Send**

Docs | Params | Authorization | Headers (9) | Body | Scripts | Settings | Cookies

none | form-data | x-www-form-urlencoded | raw | binary | GraphQL

	Key		Value	Description
☐	file	File	Messy_data_enterprise_10k_NS...	
☐	file	File	converted_output.csv	
☑	file	File	100_rows_data_main.xlsx	
☐	file	File	test.xlsx	
	Key	Text	Value	Description

Bulk Edit

Body | Cookies | Headers (4) | Test Results

202 Accepted • 86 ms • 217 B • Save Response

JSON | Preview | Visualize


```
{
    "filename": "100_rows_data_main.xlsx",
    "job_id": "57a6bb83-5f30-5a34-b997-d6c2cf64f871"
}
```


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VIEWING STATUS

Data_cleaner / **status**

Save | Share

GET http://127.0.0.1:8000/status/57a6bb83-5f30-5a34-b997-d6c2cf64f871 Send

Docs | Params | Authorization | Headers (7) | Body | Scripts | Settings | Cookies

Auth Type: Bearer Token

Token: {{token}}

The authorization header will be automatically generated when you send the request. Learn more about Bearer Token authorization.

Body | Cookies | Headers (4) | Test Results

200 OK • 38 ms • 465 B • Save Response

JSON | Preview | Visualize


```
{
    "job_id": "57a6bb83-5f30-5a34-b997-d6c2cf64f871",
    "status": "done",
    "created_at": "2026-03-07T09:25:36.381676+05:30",
    "updated_at": "2026-03-07T09:25:40.488747+05:30",
    "finished_at": "2026-03-07T09:25:40.488747+05:30",
    "download_cleaned": "/download/57a6bb83-5f30-5a34-b997-d6c2cf64f871",
    "download_logs": "/logs/57a6bb83-5f30-5a34-b997-d6c2cf64f871"
}
```


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DOWNLOADING CLEANED FILE

Data_cleaner / **Download**

Save | Share

GET http://127.0.0.1:8000/download/57a6bb83-5f30-5a34-b997-d6c2cf64f871 Send

Docs | Params | Authorization | Headers (7) | Body | Scripts | Settings | Cookies

Auth Type: Be...

Token: {{token}}

The authorization header will be automatically generated when you send the request. Learn more about Bearer Token authorization.

Body | Cookies | Headers (8) | Test Results

200 OK • 13.67 KB • Save Response

UTKARSH RANJAN

DOWNLOADING LOGS FILE

Data_cleaner / **Logs**

Save | Share

GET http://127.0.0.1:8000/logs/57a6bb83-5f30-5a34-b997-d6c2cf64f871 Send

Docs | Params | Authorization | Headers (7) | Body | Scripts | Settings | Cookies

Auth Type: Be...

Token: {{token}}

The authorization header will be automatically generated when you send the request. Learn more about Bearer Token authorization.

Body | Cookies | Headers (8) | Test Results

200 OK • 13.71 KB • Save Response

USER DATABASE TABLE

Showing rows: 1 to 2 Page No: 1 of

	id [PK] integer	email character varying	password character varying	Time timestamp with time zone
1	1	abc11@gmail.com	$2b$12$N3LgyUXLyL.2SImMqMiEdO1d3ckU3Am31emAPaswg/YNeHZ...	2026-02-05 20:24:08.560716+05:...
2	2	jonsnow21@gmail.c...	$2b$12$1hMHrp1br0pVfflkczeXhuS7mysAB2gbEXuE0l2O4E/IWW6zH8...	2026-02-05 20:24:51.121187+05:...

AUDIT JOB DATABASE TABLE

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	job_id [PK] character varying	user_email character varying	filename character varying	file_size_bytes bigint	mime_type character varying
1	dd9b897f-86bc-5d32-bfca-667fb48289...	jonsnow21@gmail.c...	converted_output.csv	14229	text/csv
2	89865851-4d78-5ea1-959d-8abcf634e...	jonsnow21@gmail.c...	test.xlsx	14974	application/vnd.openxmlformats-officedocument.spreadsheetml.s...
3	57a6bb83-5f30-5a34-b997-d6c2cf64f8...	jonsnow21@gmail.c...	100_rows_data_main.xl...	20772	application/vnd.openxmlformats-officedocument.spreadsheetml.s...

	final_status character varying	downloaded_cleaned boolean	downloaded_logs boolean	created_at timestamp with time zone	finished_at timestamp with time zone	updated_at timestamp with time zone
1	done	true	true	2026-02-28 20:43:19.126601+05:...	2026-03-01 22:10:56.819844+05:...	2026-03-01 22:11:34.464999+05:...
2	done	true	true	2026-03-04 09:25:36.758301+05:...	2026-03-04 09:25:45.127032+05:...	2026-03-04 09:26:29.386783+05:...
3	done	true	true	2026-03-07 09:25:36.381676+05:...	2026-03-07 09:25:40.488747+05:...	2026-03-07 09:26:00.326335+05:...

	attempts integer	input_path character varying	output_path character varying
1	2	[redacted]Temp\upload_olyyw6ou.c...	[redacted]\Temp\cleaned_4mj_cqzr.c...
2	1	[redacted]Temp\upload_cxwqfbq4....	[redacted]Temp\cleaned_p_mwiq3m...
3	1	[redacted]Temp\upload_7b1yxgp3....	[redacted]Temp\cleaned_34809nee....

	log_path character varying	date_intelligence_switch boolean	version character varying	version_hash character varying
1	[redacted]\Temp\log_25hcexoj.c...	true	1.0.0	98661ac158b46b8164d7c1d84d68909d255b46846cfaaba56542d9fe4e38
2	[redacted]\Temp\log_4yho5f6u....	false	1.0.1	f292619d6512a23bd6d39c3bff1b518fd3312327f1780570cdd58a962d8de
3	[redacted]\Temp\log_pnf_6cde....	true	1.0.0	98661ac158b46b8164d7c1d84d68909d255b46846cfaaba56542d9fe4e38

Conclusion

The Intelligent Data Cleaning Bot enables:

- Workforce time reduction
- Scalable concurrent processing
- Structured audit trace
- Deterministic data standardization
- Secure internal deployment

It transforms raw datasets into analytics-ready structured outputs while maintaining human oversight for flagged cases. It has been designed with the view of keeping Human in the loop concept for final checking and targeted intervention. The system performs most of the deterministic cleaning tasks automatically, allowing human reviewers to focus only on flagged edge cases thus reducing manual effort significantly, allowing the employees to handle greater number of files per day, save more time and send larger amount of data within lesser amount of time for exploratory analysis enhancing robust and faster decision making.

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